

Expected-Information-Gain-Guided Bayesian Calibration of Magnetic-Core Loss and Permeability Models for Power Converters

Viet Hoang Duong¹, Viet Huy Duong², Lun-Min Shih¹

¹Department of Computer Science, Da-Yeh University, Taiwan

²Department of Computer Science, Kent State University, United States of America

Abstract

Magnetic-core models used in power-converter design are commonly fitted to nominal material curves despite limited operating-point coverage and material variation. We present a Bayesian framework that calibrates Steinmetz core-loss and Cole–Cole complex-permeability models and evaluates expected information gain (EIG) for sequential measurement selection. Five prior-predictive, matched-model recovery cases gave per-parameter median absolute relative errors from 0.27% to 5.85%, with the generating values inside 28 of 30 equal-tailed 90% model-conditional credible intervals. In a separate 30-paired-seed benchmark with shared candidate outcomes, raw EIG reached a prespecified two-target latent-response precision gate in five measurements, compared with nine for a deterministic fixed channel-balanced traversal. Against stronger comparators, however, raw EIG tied predictive variance and Laplace D-optimality on measurement count in all 30 pairs. EIG per modeled cost tied Laplace D-optimality but required a mean 15.17 more modeled-cost units than predictive variance (95% paired-bootstrap interval 15.00–15.50 units in favor of predictive variance). No policy failed the gate. On accepted public measured records, in-sample RRMSE was 8.79%–18.21% for core loss, 6.89%–9.33% for μ' , and 36.77%–52.42% for μ'' ; the loss-component residuals expose substantial one-pole model discrepancy. The evidence supports a reproducible, model-conditional acquisition benchmark. A separate preregistered 120-task synthetic mismatch campaign found that raw EIG reached the same local gate in every combined-mismatch seed but was false-confident in 22 of 30; its strong-comparator count advantages were only 0.10 measurements. This shows that local posterior width did not protect truth accuracy under the locked departures. It does not show EIG superiority over strong comparators, measured laboratory-time savings, global six-parameter identification, or a validated optimal laboratory plan. Evidence E2, E4, E7, and E16

Keywords: Bayesian calibration; magnetic core; core loss; complex permeability; sequential experimental design; expected information gain.

I. Introduction

Power-converter efficiency, thermal margin, and magnetic-component sizing depend on models for volumetric core loss and complex permeability. Vendor curves provide essential design information, but cover a finite set of frequencies, flux densities, and temperatures and normally do not provide a joint parameter distribution. Fitting one nominal curve can therefore hide parameter non-identifiability and model discrepancy.

Bayesian calibration represents uncertainty in model parameters conditional on the selected model, observations, and prior. Bayesian experimental design ranks a candidate measurement by its expected reduction of posterior uncertainty [1], [2]. Here this framework is implemented as greedy one-step EIG over a finite library, not a globally optimal sequential policy. This is useful when characterization time is limited and dense uniform sweeps are costly. It also makes the measurement objective explicit: the selected point should reduce uncertainty in the quantities represented by the model, not merely occupy an unmeasured location in a frequency–flux grid.

This work contributes (1) a reproducible posterior formulation for Steinmetz and Cole–Cole parameters, (2) a Fisher-spectrum diagnosis of locally constrained parameter combinations, and (3) an eight-policy paired benchmark covering EIG, fixed and randomized channel-balanced traversal, predictive variance, and Laplace D-optimality under raw and modeled-cost objectives. The evidence also exposes estimator convergence, adaptive MCMC diagnostics at every acquisition state, paired candidate outcomes, disjoint holdout prediction, six-parameter recovery, measured-data acceptance gates, the raw-to-aggregate evidence chain, and a

preregistered synthetic model- mismatch evaluation. These elements answer distinct questions and cannot be substituted for one another.

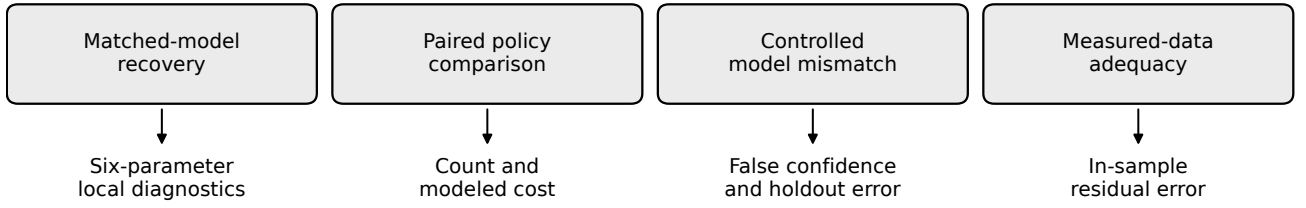
The remainder of the paper reviews magnetic and Bayesian design literature, defines the forward and statistical models, documents the reproducible protocols, and then reports matched-model, structural-mismatch, and measured-data results with their distinct evaluation scopes.

II. Research Questions and Evidence Scope

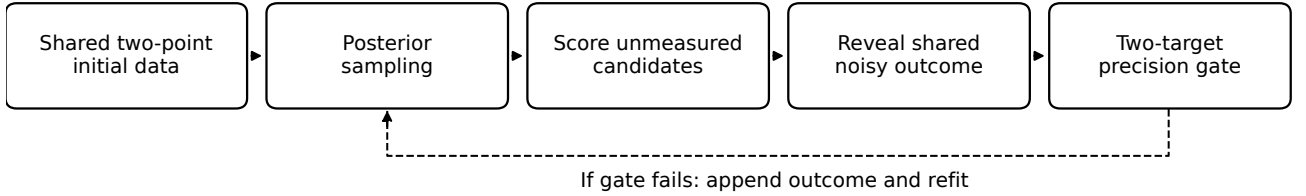
The study is organized around six research questions. *RQ1* asks whether the implemented six-coordinate inference problem is locally sensitive in all active directions and whether the posterior computation can recover known matched-model parameters. *RQ2* asks whether greedy raw EIG reaches one declared local precision gate with fewer measurements than predictive variance and Laplace D-optimality when every policy sees identical candidate outcomes. *RQ3* asks the corresponding question for EIG per prespecified modeled cost. *RQ4* asks whether the final posterior predicts a disjoint 23-point latent holdout and recovers all six generating parameters. *RQ5* asks how well the low-order forward laws describe accepted public measured records in sample. *RQ6* asks whether the same local precision gate remains truth-accurate under three fixed structural departures and whether EIG separates from strong acquisition comparators there.

The evidence ladder shown above prevents answers from being combined beyond their design. Local Fisher rank is not global identifiability; matched-model recovery is not

(a) Separate evidence streams



(b) Shared-outcome acquisition loop



(c) Scientific scope

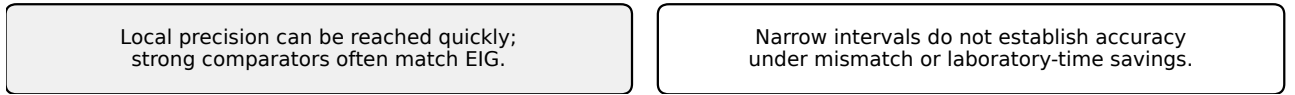


Fig. 1. Study structure separating matched-model recovery, acquisition-policy comparison, controlled structural mismatch, and measured-data adequacy.

validation on real ferrites; posterior precision is not truth proximity; and in-sample measured residuals are not held-out predictive accuracy. The acquisition result is therefore an algorithmic, model-conditional comparison under a finite library and a local gate. This explicit estimand is narrower than “optimal magnetic-component identification,” but it is reproducible and falsifiable.

The current frozen evidence contains eight policies: raw EIG, EIG divided by modeled cost, deterministic and randomized channel-balanced traversal, predictive variance under raw and per-cost objectives, and Laplace D-optimality under raw and per-cost objectives. The four direct strong-comparator contrasts were declared before the confirmatory seed results were aggregated. The fixed and randomized traversals remain useful contextual baselines, but they are not evidence that EIG dominates modern acquisition heuristics. Campaign source E1

III. Related Work

3.1 Magnetic loss and permeability models

The Steinmetz power law remains a compact engineering description for sinusoidal loss, while loss-separation and hysteresis models provide more physical decompositions [3], [4], [5]. Converter waveforms motivated modified and generalized Steinmetz relations that integrate the instantaneous flux-density trajectory or use waveform correction factors [6], [7], [8]. Measurement-led and loss-separation approaches further show why coefficients fitted under one waveform or operating window need not transfer unchanged to another [9], [10]. Subsequent work improved practical loss calculation and reviewed the physical contributions to

loss in soft ferrites [11], [12]. Textbook treatments emphasize that material law, geometry, winding, and thermal assumptions must remain distinct in component design [13], [14].

Frequency-dispersive permeability is commonly described through relaxation or resonance spectra. The Cole–Cole law introduced a broadened relaxation distribution [15]; applying its algebraic form to permeability is an engineering adaptation rather than a claim that ferrite magnetization is a dielectric process. Classical ferrite dispersion theory and measurements of Mn–Zn and Ni–Zn ferrites show that domain-wall motion, rotational processes, resonance, and relaxation can all shape μ' and μ'' [16], [17]. The single-pole model used here is therefore intentionally diagnostic: a large loss-component residual is evidence for missing dynamics, not a parameter-estimation failure alone.

Open magnetic datasets now permit reproducible comparisons beyond individual vendor plots. MagNet provides systematically measured core-loss data across materials and excitations [18], while the LEA material database publishes machine-readable permeability and loss records with source metadata [19]. Public data improve traceability but do not substitute for a controlled multi-lot characterization campaign.

3.2 Bayesian calibration and experimental design

Bayesian calibration represents parameter uncertainty conditional on a forward model, likelihood, and prior [20]. Computer-experiment literature distinguishes emulator uncertainty, observation error, calibration parameters, and structural discrepancy [21], [22]. Validation frameworks conse-

quently compare predictions with observations rather than treating posterior concentration as proof of physical validity [23]. Ignoring discrepancy can bias inferred physical parameters and produce overconfident extrapolation [24]. Calibration theory for imperfect simulators also demonstrates that the physical parameter and discrepancy can be confounded unless additional structure is imposed [25], [26]. We do not introduce an unidentified discrepancy process into the present six-parameter fit; instead, measured residuals are reported as a separate adequacy diagnostic.

Bayesian experimental design formalizes the value of a proposed observation through expected utility. Lindley's information criterion [27], foundational reviews [1], [2], and classical optimum-design methods [28] motivate the EIG criterion used here. Estimation remains challenging because the predictive density is generally unavailable; relevant strategies include Laplace and MCMC approximations [29], nested Monte Carlo [30], multilevel estimators [31], stochastic simulation-based optimization [32], and variational bounds [33]. Adaptive-policy amortization offers a scalable direction for longer experimental sequences [34], but the present work deliberately uses a transparent one-step ranking over a finite candidate library.

Information-based acquisition connects Shannon entropy and Kullback–Leibler divergence to active data selection [35], [36], [37]. Classical D-optimal design instead optimizes a determinant of an information matrix and is linked to equivalence theorems [28], [38]. These criteria coincide only under particular approximations; the fixed traversal used in the reported benchmark is neither of them.

IV. Magnetic-Core Models

4.1 Core loss

For sinusoidal excitation within a specified operating window, the Steinmetz model is

$$P_v(f, B_{pk}) = k f^\alpha B_{pk}^\beta,$$

where P_v is volumetric loss and (k, α, β) are inferred parameters [3]. The units of k depend on the units used for P_v , f , and B_{pk} . The implemented Steinmetz equation is not temperature dependent in the present implementation. Measured-data analysis therefore uses a specified temperature subset; the structural-mismatch experiment separately tests the consequences of extrapolating this isothermal model across temperature. More general waveform-dependent loss equations remain important for converter application studies [6], [11].

4.2 Complex permeability

We adopt the $e^{j2\pi ft}$ time convention and write $\mu_r = \mu' - j\mu''$, so the reported loss component is $\mu'' = -\text{Im}\mu_r \geq 0$ for a passive response. The relative complex permeability is represented with a single Cole–Cole relaxation [15],

$$\mu_r(f) = 1 + \frac{\mu_s - 1}{1 + (jf/f_{rel})^{1-a_{cc}}},$$

where μ_s is low-frequency permeability, f_{rel} is a relaxation frequency, and a_{cc} broadens the relaxation. Storage and loss components are evaluated separately because a model can fit μ' while failing to represent μ'' . A one-pole model is a deliberate low-order approximation, not a complete ferrite loss model.

The implemented Cole–Cole equation fixes the high-frequency asymptote to unity and absorbs the angular-frequency convention into f_{rel} . The real and loss components follow from the same complex denominator, which prevents independent tuning of μ' and μ'' . For an optional inductance channel, the real component is mapped through $L_m = \mu_0 \mu' N^2 A_e / \ell_e$ using specified geometry [13], [14]. Geometry is never inferred jointly with material parameters in the present study.

4.3 Validity window

Both forward laws are isothermal and evaluated only inside specified data ranges. A data point is defined by channel, frequency, peak flux density, and temperature. Core-loss points require positive B_{pk} ; permeability points use small-signal records. The likelihood rejects mixed temperature cohorts. Measured permeability uses material-specific cohorts of $25 \pm 0.75^\circ\text{C}$ for N87 and $30 \pm 0.75^\circ\text{C}$ for N95. Measured core loss uses $25 \pm 0.5^\circ\text{C}$ for N49, N87, and N95, and $100 \pm 0.5^\circ\text{C}$ for 3C95. These restrictions prevent an apparently precise posterior from silently extrapolating across temperature, waveform, or excitation regime.

V. Bayesian Calibration and Design

Let $\theta = (\log k, \alpha, \beta, \log \mu_s, \log f_{rel}, a_{cc})$ denote the active transformed parameters. Log coordinates enforce positivity of scale parameters while retaining direct coordinates for the exponents. For observation y_o at design d_o , the likelihood is

$$p(\mathbf{y} | \theta, \mathbf{d}) = \prod_o \mathcal{N}(y_o; F_o(\theta, d_o), \sigma_o^2),$$

with fixed relative scales. Synthetic scales are 3% for core loss, 2% for μ' , 5% for μ'' , and 1% for inductance; measured records use a 3% relative scale as a weighting model. The posterior is

$$p(\theta | \mathbf{y}, \mathbf{d}) \propto p(\mathbf{y} | \theta, \mathbf{d}) p(\theta).$$

Priors are defined directly in the transformed space used by the sampler, so no inverse-transform Jacobian is added to this active-coordinate density. Bounds encode physically admissible numerical domains, not empirical proof that the chosen model is correct.

The six coordinates divide into two response blocks in the likelihood but are sampled jointly. Core-loss observations directly update $(\log k, \alpha, \beta)$, while permeability and inductance observations update $(\log \mu_s, \log f_{rel}, a_{cc})$. Joint sampling retains a single posterior contract and permits one acquisition library to contain all channels; it does not create physical coupling absent from the forward laws. The parameter table makes this structural separation explicit.

Four uncertainty sources must be distinguished. *Parameter uncertainty* is represented by posterior draws conditional

TABLE I
Physical parameters and active inference coordinates.

Physical parameter	Active coordinate	Role	Principal response	Main limitation
k	$\log k$	core-loss scale	P_v	units follow the declared convention
α	α	frequency exponent	P_v	local to sinusoidal, isothermal data
β	β	flux-density exponent	P_v	no waveform or DC-bias dependence
μ_s	$\log \mu_s$	low-frequency permeability	μ', L_m	fixed rather than inferred geometry
f_{rel}	$\log f_{\text{rel}}$	relaxation frequency	μ', μ''	one effective relaxation only
a_{cc}	a_{cc}	relaxation broadening	μ', μ''	no multiple resonances

on the chosen model and fixed geometry. *Observation uncertainty* enters through the declared Gaussian scale. *Monte Carlo uncertainty* arises because posterior and EIG integrals are approximated by finite samples. *Structural uncertainty* is exposed by measured residuals but is not parameterized in the likelihood. Only the first three are quantified in parts of the present workflow; an EIG score interval is therefore not a total physical-uncertainty interval.

5.1 Posterior computation and diagnostics

Posterior sampling uses the affine-invariant ensemble construction [39] implemented by `emcee` [40]. Recovery and measured-data fits retain 5000 steps after 1000 warm-up steps for each of 48 walkers. Matched-model acquisition states use a stricter adaptive contract: 4000 warm-up steps, at least 20,000 retained steps, checks every 10,000 additional steps, and a hard ceiling of 80,000 retained steps. The same state-keyed chain is extended rather than restarted. Initial states are perturbations of the fixed prior center, never the hidden generating truth. A state is accepted only with finite integrated autocorrelation estimates, at least 50 retained steps per autocorrelation time, effective sample size at least 400, mean acceptance between 0.20 and 0.60, and finite log probability for every retained sample. Failure at the hard ceiling invalidates that policy record.

These checks follow the principle of comparing Monte Carlo variation with between-sequence behavior [41] and modern effective-sample-size practice [42], but interacting ensemble walkers are not treated as independent chains. Accordingly, the paper does not report a conventional $\hat{R}$ for the ensemble. Diagnostic failure invalidates a measured fit for quantitative publication rather than being repaired by discarding an unfavorable seed.

Matched-model recovery also exercises the complete forward–likelihood–sampler–summary path against known generating values. It is an implementation check related to posterior-quantile validation [43], but five independent fits do not constitute simulation-based calibration [44]. A calibrated rank experiment would require many more prior-predictive replications and a uniform-rank diagnostic. Posterior and residual graphics are interpreted as part of a Bayesian workflow rather than as substitutes for numerical diagnostics [45], [46].

The local Fisher approximation [47] is

$$\mathcal{J}_{ij} = \sum_o \frac{1}{\sigma_o^2} \frac{\partial F_o}{\partial \theta_i} \frac{\partial F_o}{\partial \theta_j}.$$

Its spectrum is computed over the exact active vector and with reported scaling. A large condition number indicates locally sloppy combinations but does not by itself establish global non-identifiability [48]. Geometric analyses of sloppy models likewise interpret broad eigenvalue spectra as anisotropic distinguishability, not as a universal parameter ranking [49]. The spectrum is therefore reported together with its dimension and scaling convention.

Let $\mathcal{D}_t = (\mathbf{y}_t, \mathbf{d}_t)$ denote the observations available before acquisition step t . For candidate d , the one-step EIG is the conditional mutual information [1], [27]

$$\text{EIG}_t(d) = \mathbb{E}_{\substack{\theta \sim p(\theta | \mathcal{D}_t) \\ y \sim p(y | \theta, d)}} [\log p(y | \theta, d) - \log p(y | d, \mathcal{D}_t)],$$

where

$$p(y | d, \mathcal{D}_t) = \int p(y | \vartheta, d) p(\vartheta | \mathcal{D}_t) d\vartheta.$$

For outer draws (θ_i, y_i) and inner posterior draws ϑ_{ij} , the implemented nested estimator is

$$\hat{U}(d) = \frac{1}{N} \sum_{i=1}^N \left[\log p(y_i | \theta_i, d) - \log \left\{ \frac{1}{M} \sum_{j=1}^M p(y_i | \vartheta_{ij}, d) \right\} \right].$$

Finite inner budgets introduce bias through the logarithm and finite outer budgets introduce variance; close candidates can therefore exchange rank [30], [31]. The convergence campaign described below tests ranking and downstream endpoint stability rather than assuming one budget is sufficient.

The reported estimator draws outer parameter–observation pairs and uses inner posterior draws to approximate the posterior-predictive density [29], [30]. Each score is repeated 40 times; the mean, standard error, 95% Monte Carlo interval, and top-rank frequency are retained. Candidate streams are seeded from a stable key comprising channel, frequency, flux density, and temperature, so list permutation

cannot change a design’s draws. We evaluate two distinct objectives: raw EIG for the measurement-count endpoint and EIG divided by a prespecified modeled acquisition cost for the total-cost endpoint. These intervals quantify repeated nested-Monte-Carlo variability conditional on the retained posterior draws and model; they exclude chain-to-chain, structural-model, geometry, and noise-model uncertainty.

5.2 Paired sequential policy

After the same two initial observations— P_v at 30 kHz, 0.05 T, and 25°C, and μ' at 10 kHz and 25°C—each policy selects one unrevealed candidate, receives its pre-generated noisy outcome, refits the posterior, and checks the common precision gate. The registry contains exactly eight policies. Raw EIG and predictive variance maximize their unscaled criteria; their per-cost forms divide those criteria by the declared channel cost. Laplace D-optimality greedily maximizes the log-determinant increment of a local posterior-precision approximation, with an analogous per-cost form. The contextual comparators are deterministic fixed and seeded randomized channel-balanced traversals. The fixed traversal is not called a uniform grid, and neither traversal is a space-filling or globally optimal design. All policies share the candidate library, latent truth, candidate-indexed noisy outcomes, maximum budget, and stopping rule. The candidate library is isothermal, so it contains no temperature-only duplicates of the implemented forward laws. The numerical settings are summarized below.

For clarity, predictive variance is noise-normalized, not an unscaled comparison of responses with different physical units. With posterior response draws $F_d(\theta)$, its score is

$$U_{PV}(d) = \frac{\text{Var}[F_d(\theta) | \mathcal{D}_t]}{\sigma_d^2}.$$

The denominator uses the declared channel noise fraction times the median absolute posterior prediction, with the implementation’s 10^{-9} numerical floor. Irreducible observation variance is not added to the numerator. For posterior covariance Σ in active coordinates and response gradient g_d at the posterior-mean coordinate vector, the Laplace score is

$$U_L(d) = \frac{1}{2} \log \left(1 + \frac{g_d^T \Sigma g_d}{\sigma_d^2} \right).$$

It uses central finite differences with relative step 10^{-5} and the same noise convention. This is a local Gaussian information approximation rather than a full nonlinear Fisher-greedy policy. In the linear-Gaussian limit, the raw Laplace score is a monotone function of noise-normalized predictive variance, explaining why their rankings can agree. Dividing the two scores by varying channel costs need not preserve this agreement.

The stopping statistic is the half-width of the central 90% posterior interval for the noise-free latent mean response divided by the absolute posterior median. Observation noise is not added to this gate. The two fixed targets are $P_v(100\text{kHz}, 0.1\text{ T}, 25^\circ\text{C})$ at 8% and $L_m(100\text{kHz}, 25^\circ\text{C})$ at 5%. This is a local precision criterion: it checks neither truth proximity, six-parameter recovery, other channels, held-out

performance, nor model adequacy. A policy that misses either threshold is recorded as a failure at the maximum budget. Candidate EIG integrates over the joint six-parameter posterior; only the stopping and evaluation endpoint is restricted to these two responses.

All 37 candidates are at 25°C. Modeled costs are positive prespecified units used for the modeled-cost objective and endpoint; they are not measured instrument or laboratory durations. The two initial observations belong to the same library and count toward the stopping total.

VI. Computational Reproducibility

All experiments use fixed software versions, specified model and inference settings, explicit data selections, and deterministic random seeds. Results are retained at the individual-seed level and are accepted for aggregation only when all prespecified validity gates pass.

The analysis checks all required seed and material cases before forming aggregate summaries. Records with nonfinite values, failed convergence, or boundary-active parameters are excluded according to criteria specified before evaluation. Numerical statements, tables, and figures are generated programmatically from the same complete set of accepted records, avoiding manual transcription or selective replacement of outcomes.

6.1 Estimator-budget qualification

The nested estimator was qualified before the 30-seed acquisition campaign. Twelve posterior states were formed from four seeds (7100–7103) after 2, 4, and 6 observations. At each state, 27 candidate settings crossed outer budgets {100, 300, 900}, inner budgets {50, 100, 300}, and replicate counts {5, 10, 20}. Every setting was compared with a 1200-outer, 400-inner, 40-replicate reference. Two sentinel states were additionally recomputed at 2400 outer and 800 inner draws with 40 replicates. Prefix-nested random streams made the lower-budget samples literal prefixes of the larger calculations, reducing irrelevant differences between settings.

The acceptance decision combined the stable top candidate, Spearman rank correlation, score-interval overlap, relative reference regret, and exact agreement of downstream count and modeled-cost endpoints. Ten disjoint seeds (7200–7209) supplied that downstream check. The selected setting was then locked before the confirmatory acquisition seeds 7300–7329 were evaluated. This separation prevents selecting an estimator budget because it happened to produce the most favorable headline comparison.

The public audit projection retains the campaign records needed to reconstruct the acquisition endpoints and estimator decision while removing machine paths and scheduler metadata. Posterior arrays are retained flattened over sampling draws; because walker and iteration indices are not present, they are not described as raw MCMC chains. The compact frozen summary authenticates the reported measured-fit aggregates and exclusions, but the audit projection does not independently reconstruct those fits. Raw upstream measured curves remain governed by their source repositories

TABLE II
Isothermal acquisition library, observation scales, and modeled costs.

Channel	Frequencies (kHz)	B_{pk} (T)	Points	Relative noise	Modeled cost
P_v	30, 50, 100, 200, 300, 500	0.05, 0.10, 0.20	18	3%	60
μ'	10, 30, 100, 300, 500, 1000, 2000, 3000	0	8	2%	20
μ''	10, 30, 100, 300, 500, 1000, 2000, 3000	0	8	5%	20
L_m	10, 100, 1000	0	3	1%	15

TABLE III

Computational settings for the matched-model acquisition experiment. The structural-mismatch sampler is specified separately below.

Item	Setting
Ensemble sampler	48 walkers; acquisition chains retain 20,000–80,000 adaptively
Convergence criteria	$ESS \geq 400$; $steps/\tau \geq 50$
Acceptance criterion	0.20–0.60
Nested EIG	1200 outer; 400 inner; 40 score replicates
Sequential budget	at most 25 measurements
Acquisition endpoints	raw EIG/count; EIG/cost/modelled cost
Temperature cohorts	material and channel specific, as stated in text
Precision gate	latent mean response: 8% P_v ; 5% L_m

and are not redistributed here.

VII. Evaluation Protocol

Five fixed seeds are used for matched-model recovery. A disjoint, prespecified set of seeds is used for the paired acquisition comparison only after the nested-Monte-Carlo setting passes a separate convergence study. That study uses prefix-nested common random numbers and a deterministic paired bootstrap of the replicate-mean selector; it requires stable candidate and reference winners, rank correlation, score-interval overlap, low reference regret, and downstream endpoint agreement. The policies share initial observations, candidate library, realized candidate outcomes, noise policy, budget, and stopping rule. Results include per-seed counts, modeled costs, and failures, not only an aggregate ratio. The stopping rule concerns posterior precision of two latent mean responses and is not an accuracy or coverage guarantee.

For recovery, observations are generated from the same six-parameter forward family used in inference. A parameter vector is drawn independently for each seed from a datasheet prior fixed before that draw; the prior is never rebuilt from the realized truth. Sampler walkers are initialized around the fixed prior center, not around the hidden generating vector. Observation noise is then generated independently. This is a prior-predictive matched-model design: it removes oracle centering while retaining favorable equality between the data-generating and inference model families. The reported error is the absolute relative difference between posterior median and the generating value. The interval statistic counts whether that value falls within the equal-tailed 90% model-conditional posterior credible interval. With only five seeds, the count is a descriptive implementation check, not a coverage experiment. The Fisher matrix is evaluated in the exact

transformed coordinate system used by sampling.

For the paired acquisition analysis, a stable design-keyed seed generates one noisy observation for every candidate before any policy is run. A policy reveals the already generated value when it selects that design. Thus policy order cannot change the noise attached to a candidate, and all policies in a seed share the same latent truth and candidate outcome map. The 30 paired differences are the unit of analysis. Descriptive means, medians, standard deviations, win and failure rates, and percentile intervals are computed from those differences. The reported confidence interval uses 10,000 deterministic paired bootstrap resamples. It quantifies between-seed variation in this matched-model experiment and is not a population-wide material claim.

The measured evaluation uses public complex-permeability curves for specified N95, N87, and 3C95 source/material pairs and datasheet core-loss curves for N49, N87, N95, and 3C95 [19]. Input-data versions, temperature cohorts, unit conversions, and range filters were fixed before inference. Storage/loss permeability errors, core-loss residuals, boundary activity, and operating-window coverage are reported separately. Records with failed convergence or active boundary flags are retained for diagnostic review but excluded from aggregate numerical claims.

Core-loss adequacy is summarized by in-sample relative root-mean-square error (RRMSE) within each temperature cohort. Permeability adequacy is evaluated separately for storage and loss components because combining them can conceal a poor μ'' fit behind the larger magnitude of μ' . These metrics assess the selected low-order model on observed records; they are neither held-out prediction errors nor component-level thermal validation.

TABLE IV
Evidence layers of the matched-model and measured-data evaluation.

Layer	Prespecified cases	Retained public evidence	Scientific role
Matched recovery	5 generating seeds	seed-level parameter summaries	implementation check
Estimator states	$4 \times 3 = 12$	state records and flattened posterior samples	budget qualification
Estimator grid	27 budget combinations across 12 states	108 candidate-score records plus 12 references	ranking stability
Doubled-budget audit	2 sentinel states	2 doubled-budget score records	reference sensitivity
Downstream validation	10 seeds	10 policy endpoint records	endpoint stability
Confirmatory acquisition	30 paired seeds	30 complete policy trajectories	count and cost endpoints
Measured adequacy	declared material/source pairs	aggregate metrics and disclosed exclusions	model discrepancy diagnostic

7.1 Controlled structural-mismatch experiment

The separate MM-3 experiment holds the inference family fixed while changing the data generator. Thirty independent prior-predictive seeds, 10100–10129, are crossed with four prespecified scenarios. Within each of the resulting 120 tasks, all eight policies share the generating parameter vector and the candidate-indexed noisy outcomes. The scenarios are a matched control, two-pole permeability, temperature/curvature core loss, and a combined departure. These are controlled perturbations, not fits of a discrepancy law to the measured residuals. The combined case has its own larger coefficients; it is not simply the sum of the two single-departure cases.

For the loss generator, define $x = \log_{10}(f/100\text{kHz})$ and $z = \log_{10}(B_{\text{pk}}/0.1\text{T})$. The generated latent loss is

$$P_v^* = k f^\alpha B_{\text{pk}}^\beta \exp\{s_T(T - 25) + c_f x^2 + c_{fB} x z\}.$$

Here T is the numerical temperature in degrees Celsius and s_T has units of inverse degrees Celsius. The two-pole permeability generator is

$$\mu_r^*(f) = 1 + (\mu_s - 1) \sum_{r=1}^2 \frac{w_r}{1 + [j f / (m_r f_{\text{rel}})]^{1-\tilde{a}_r}},$$

where $w_1 = 1 - w_2$ and $\tilde{a}_r = \min\{0.95, \max[0, a_{\text{cc}} + \delta_r]\}$. The inductance generator uses the real part of this permeability and the same fixed geometry as inference. The registered coefficients are shown below; none is inferred by the six-parameter posterior.

Acquisition retains exactly the 37 unique 25°C candidates and the original two-target precision gate. Temperatures of 60 and 100°C occur only in the disjoint latent holdout: replicating temperature-only candidates would add duplicate information under the isothermal inference model. The mismatch holdout contains 39 points, extending the matched-model 23-point set by repeating its eight core-loss points at those two additional temperatures. Holdout truths are never used for acquisition or stopping. Thus the cross-temperature result measures extrapolation failure, not learning from temperature measurements.

Production sampling uses an 80% differential-evolution and 20% snooker move mixture in `emcee` 3.1.6 with 48 walkers, initialized at perturbations of the fixed prior center. The differential-evolution scale is $2.38/\sqrt{12}$; the snooker

scale is 1.7. Each state discards 80,000 warm-up steps and is checked every 20,000 retained steps up to a hard limit of 800,000. Every coordinate must have ESS at least 400 and retained length divided by autocorrelation time at least 50, with finite log probability throughout and mean acceptance in $[0.05, 0.80]$. Two successive checkpoint changes in every autocorrelation estimate must be no greater than 10%, measured relative to the current estimate; hence acceptance cannot occur before 60,000 retained steps. This contract was qualified on two locked sparse-observation states using independent ensembles without acquisition endpoints, then fixed before the new seed campaign. Earlier incomplete campaigns are not pooled with MM-3.

A false-confident outcome is defined before evaluation as reaching the local width gate while the posterior-median absolute relative error exceeds 8% at the core-loss target or 5% at the inductance target. Latent-holdout RRMSE and 90% interval inclusion are reported separately by channel and temperature. The unit for paired inference remains the complete seed, and all bootstrap intervals are descriptive, without multiplicity adjustment. A 90% interval’s inclusion frequency over this fixed holdout is not a parameter-rank calibration diagnostic. Complete admission requires all 120 expected tasks and all their policy records to satisfy the locked validity checks; a partial matrix is not an alternative confirmatory result. Protocol and result source E16

VIII. Results

8.1 Identifiability and synthetic recovery

The synthetic 6×6 Fisher matrix was full rank. Its eigenvalues span 1.24×10^2 to 2.91×10^6 , giving a condition number of 2.35×10^4 . This broad spectrum indicates strong local sensitivity anisotropy in the active transformed coordinates; it does not establish global identifiability. Across five prior-predictive matched-model data sets, the per-parameter median absolute relative errors ranged from 0.27% to 5.85%. The known values fell within 28 of 30 equal-tailed 90% model-conditional posterior credible intervals. This observed inclusion count is descriptive, not a coverage estimate. Result source E2

Table source E2

TABLE V
Fixed structural-departure coefficients in the data generator.

Scenario	w_2	m_1, m_2	δ_1, δ_2	s_T	c_f	c_{fB}
Matched control	0	1, 1	0, 0	0	0	0
Two-pole permeability	0.25	0.75, 5	-0.05, 0.18	0	0	0
Temperature/curvature	0	1, 1	0, 0	0.0035	0.08	0.05
Combined mismatch	0.35	0.55, 7	-0.10, 0.25	0.0055	0.14	0.10

TABLE VI
Five-seed matched-model parameter recovery; errors are absolute relative errors.

Parameter	Mean error	Median error	SD	Truth in 90% CI
k	7.36%	5.85%	5.94%	4/5
α	0.40%	0.27%	0.36%	4/5
β	0.38%	0.36%	0.25%	5/5
μ_s	0.35%	0.38%	0.17%	5/5
f_{rel}	0.91%	0.87%	0.55%	5/5
a_{cc}	0.91%	0.62%	0.79%	5/5

8.2 Nested-estimator qualification

The conservative reference setting of 1200 outer draws, 400 inner draws, and 40 score replicates was selected by the prespecified convergence decision. Against doubled-budget sentinels, minimum top-selector agreement was 0.9983 for raw EIG and 1.0000 for EIG/cost; minimum Spearman correlation was 0.9983 and 0.9992, respectively, minimum score-interval overlap was 0.8710, and maximum relative regret was zero. Ten downstream validation seeds reproduced the reference count and modeled-cost endpoints. This qualifies the numerical setting for the campaign; it does not make each finite EIG score exact or incorporate structural-model uncertainty. Result source E3

8.3 Paired synthetic acquisition

Raw EIG reached the two-target precision gate in exactly five measurements, while the deterministic fixed channel-balanced traversal required nine in all 30 paired seeds. The four-measurement difference had zero sample variation and a bootstrap 95% interval of 4.00–4.00 measurements. This 44.4% reduction is a valid comparison with that specified baseline, but the strong-comparator results materially narrow its interpretation. Result source E4

Table source E4

All eight policies reached the gate in all 30 seeds. Thus the experiment gives no evidence that raw EIG outperforms predictive variance or Laplace D-optimality on measurement count. Under the cost objective, predictive variance reached the gate at modeled cost 175 in every seed, whereas EIG/cost used 190 in 29 seeds and 195 once. The cost constants are prespecified model inputs, not measured laboratory durations. Result source E4

8.3.1 Descriptive selection-path diagnostic

A post hoc diagnostic helps locate why the strong comparators tied or beat EIG on this gate. On posterior states with identical hashes and candidate universes, raw EIG and raw

predictive variance had mean Spearman score-rank correlation 0.9967 over 57 states; their final selected sets had mean Jaccard overlap 0.933, although the complete ordered path matched in only 11 of 30 seeds. Raw EIG and raw Laplace D-optimality similarly had mean rank correlation 0.9947 over 36 comparable states and mean selected-set Jaccard overlap 0.867. The raw utilities therefore placed nearly the same candidates near the top and usually acquired the same set in a different order, leaving no measurement-count separation at the common gate.

The cost-normalized comparison diverged at the third acquisition. EIG/cost selected $L_m(10\text{kHz})$ in 28 of 30 seeds, whereas predictive variance/cost selected $P_v(500\text{kHz}, 0.2\text{T})$ in 29 of 30. The mean reduction in the normalized maximum gate ratio was 0.0048 for EIG/cost and 2.6174 for predictive variance/cost; zero and 30 seeds, respectively, crossed the gate after that acquisition. This pattern supports objective-gate misalignment in the present matched-model benchmark: joint parameter information favored an inductance measurement while the remaining stopping uncertainty was primarily at the core-loss target. Because this diagnostic was defined after the comparator result and uses realized policy paths, it does not establish a general causal advantage for predictive variance. Diagnostic source E9

8.4 Disjoint holdout and six-parameter endpoints

The 23-point latent holdout was excluded from acquisition, policy selection, and stopping. For raw EIG, mean RRMSE was 3.12% for core loss, 1.32% for μ' , 3.93% for μ'' , and 1.20% for L_m ; mean 90% latent-interval coverage was 85.8%, 93.9%, 90.0%, and 92.2%, respectively. EIG/cost gave mean RRMSE 3.15%, 0.95%, 2.78%, and 0.85%, with coverage 85.0%, 93.3%, 89.4%, and 93.3%. These secondary endpoints do not rescue an acquisition-superiority claim: predictive-variance and Laplace policies had comparable holdout performance, while the fixed and randomized traversals were often worse. Result source E6

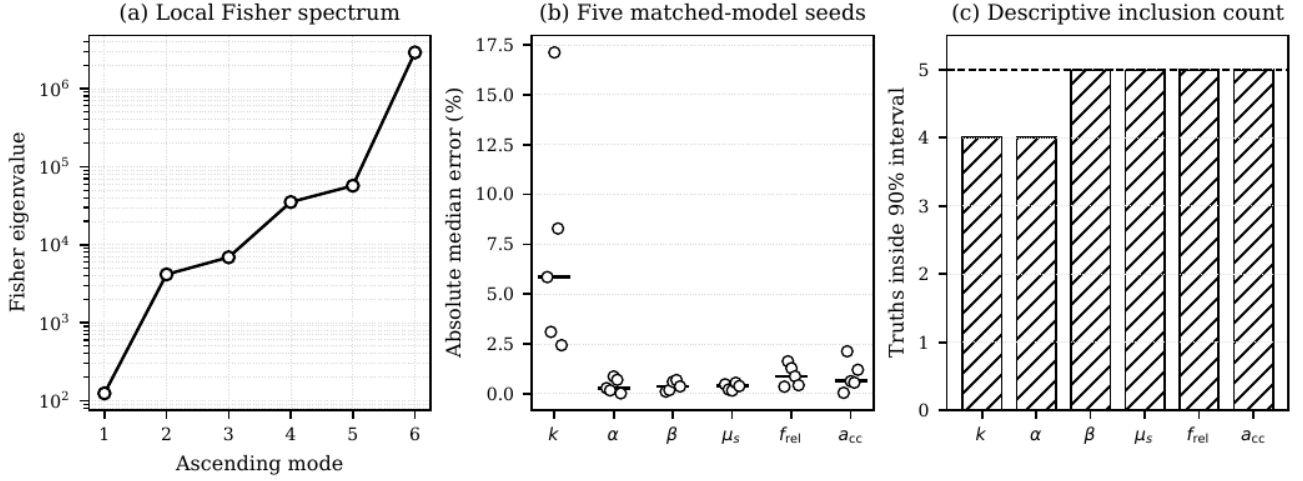


Fig. 2. Local Fisher spectrum and five-seed matched-model recovery.

TABLE VII

Direct strong-comparator contrasts over 30 paired matched-model seeds. W/T/L denotes EIG wins, ties, and losses.

Preregistered direct contrast	Endpoint	Mean difference, comparator – EIG	Bootstrap 95% CI	EIG W/T/L
Raw EIG vs predictive variance	measurement count	0.00	[0.00, 0.00]	0/30/0
Raw EIG vs Laplace D-optimality	measurement count	0.00	[0.00, 0.00]	0/30/0
EIG/cost vs predictive variance/cost	modeled cost	-15.17	[-15.50, -15.00]	0/0/30
EIG/cost vs Laplace D-optimality/cost	modeled cost	0.00	[0.00, 0.00]	0/30/0

Final 90% parameter intervals contained a mean 5.13 of six generating parameters for both EIG objectives. Across policies, however, mean absolute error for k remained approximately 25.1%–30.7%. Posterior interval inclusion must therefore not be described as uniformly precise global six-parameter recovery. Result source E6

8.5 Preregistered model-mismatch campaign

MM-3 evaluated the matched control and three fixed structural departures over 30 new paired seeds. All 120 scenario-seed tasks passed the registered sampler contract, all eight policies reached the local precision gate in every scenario, and the complete sanitized record set reconstructed the aggregate. This is a synthetic robustness experiment with latent truth, not a measured- material validation. Result source E16

The advantages are paired comparator-minus-EIG measurement counts; positive values favor EIG. Raw EIG versus predictive variance recorded 2/28/0 wins/ties/losses in each of the first three scenarios and 3/27/0 under combined mismatch. Versus Laplace D-optimality the corresponding counts were 5/25/0 and 3/27/0. Thus the observed advantages were substantially below one measurement and arose mostly from ties. EIG/cost lost to predictive-variance/cost in all 30 pairs in every scenario, by a mean 15.33 modeled-cost units in the first three and 15.17 in combined mismatch. Table source E16

Combined-mismatch raw EIG produced mean latent-holdout core-loss RRMSE of 19.68% and 90% interval inclusion of 15.69%. Loss-permeability RRMSE and inclusion were 39.76% and 35.00%; core loss at 100°C had 30.55% RRMSE and zero interval inclusion. Because the precision gate was nevertheless reached in 30/30 seeds, the gate cannot be used as a proxy for truth accuracy under these departures. The strong adaptive policies also recorded 21–23 false-confident seeds in combined mismatch. The experiment therefore does not establish that EIG uniquely causes false confidence or that either balanced traversal is generally safer. Result source E16

8.6 Measured-data model adequacy

For LEA Material Database permeability records that satisfied the convergence criteria, RRMSEs were 9.33%/52.42% for μ'/μ'' on N87 and 6.89%/36.77% on N95. The larger μ'' errors expose discrepancy in the one-pole model. N87 and 3C95 MagNet records failed convergence or boundary gates and were excluded from numerical claims [18]. Measured Steinmetz fits produced in-sample RRMSEs of 8.79% (N87), 12.55% (N49), 13.40% (3C95), and 18.21% (N95) within their specified temperature cohorts. These are fit residuals, not held-out errors. Result source E7

Each result family uses its declared models, data filters, seeds, and evaluation criteria; the mismatch sampler and

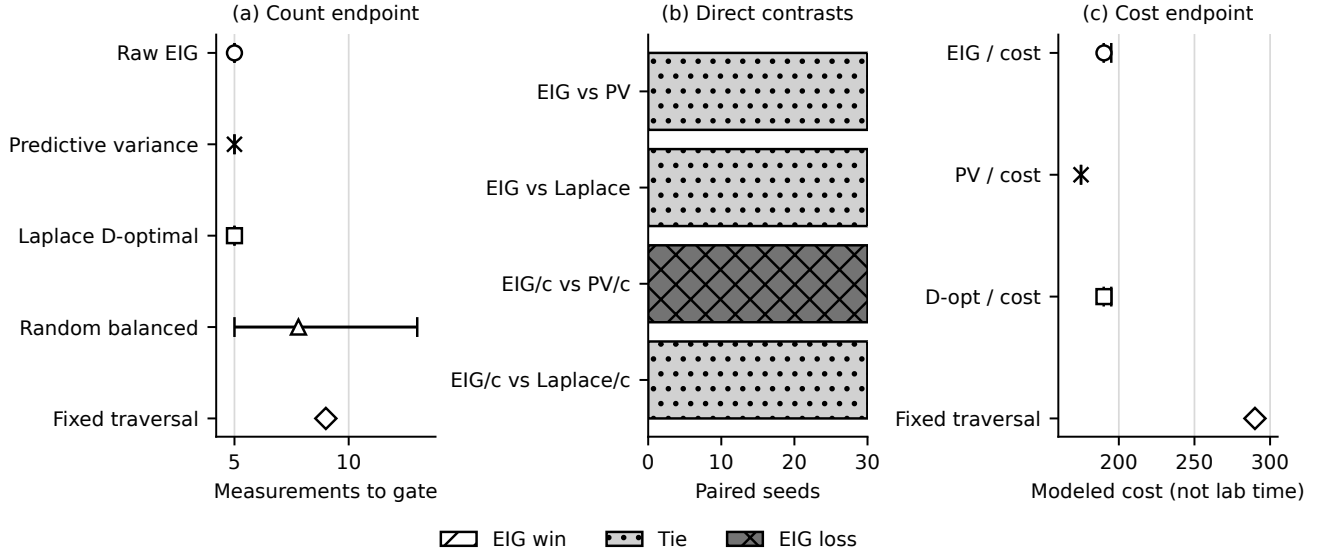


Fig. 3. Eight-policy paired acquisition results and four direct strong-comparator contrasts.

TABLE VIII
Raw EIG under the four registered data generators, with 30 paired seeds per scenario.

Scenario	Raw EIG false confidence	Mean measurements	Advantage vs predictive variance	Advantage vs Laplace D-optimality
Matched control	0/30	4.833	0.067	0.167
Two-pole permeability	9/30	4.833	0.067	0.167
Core-loss	1/30	4.833	0.067	0.167
temperature/curvature				
Combined mismatch	22/30	4.900	0.100	0.100

holdout differ explicitly from those of the earlier matched-model comparison.

IX. Discussion and Limitations

9.1 Interpretation of the evidence

The framework separates whether available measurements constrain parameters within a chosen model from whether that model represents measured material behavior. Fisher information and posterior width address the former locally or conditionally; reported in-sample residuals expose, but do not fully quantify, the latter. The disjoint synthetic holdout now measures latent prediction within the matched family; held-out validation on independently acquired physical material is not established by these results.

The synthetic results support implementation-level conclusions. Full local rank shows that every active coordinate affects the specified synthetic experiment, while the broad spectrum shows that those effects occur at very different scales. The recovery table then checks the nonlinear posterior under matched truth. Neither result implies that the same coordinates are uniquely physical when the material response lies outside the six-parameter family. This distinction matters for converter models because several parameter sets can produce similar loss or permeability over a narrow operating window yet diverge when frequency, flux density,

or temperature changes.

The paired acquisition result shows that EIG exploited posterior structure relative to the fixed traversal, but it does not isolate an EIG-specific advantage: raw predictive variance and Laplace D-optimality reached the same gate in the same number of measurements. Under the modeled-cost objective, predictive variance was better than EIG in every seed. The result means only that these policies narrow two specified latent-response intervals efficiently inside the matched model. It does not establish accurate recovery of a physical component, global predictive accuracy, or robustness to model misspecification. Contrast source E4

MM-3 sharpens this boundary. The local width gate remained easy to satisfy under the locked discrepancy terms, while truth accuracy and fixed latent- holdout inclusion deteriorated. In combined mismatch, raw EIG was false-confident in 22 of 30 seeds, yet the other strong adaptive policies produced similar counts. Model adequacy and stopping-rule design therefore affected the interpretation more than the small between-policy count differences. This is evidence about the three registered departures, not a theorem for arbitrary misspecification. Model-mismatch source E16

9.2 Why the strong comparators tie or win

The raw count tie is a resolution property of the declared stopping endpoint, not proof that the acquisition rules rank candidates identically. After four total measurements, no

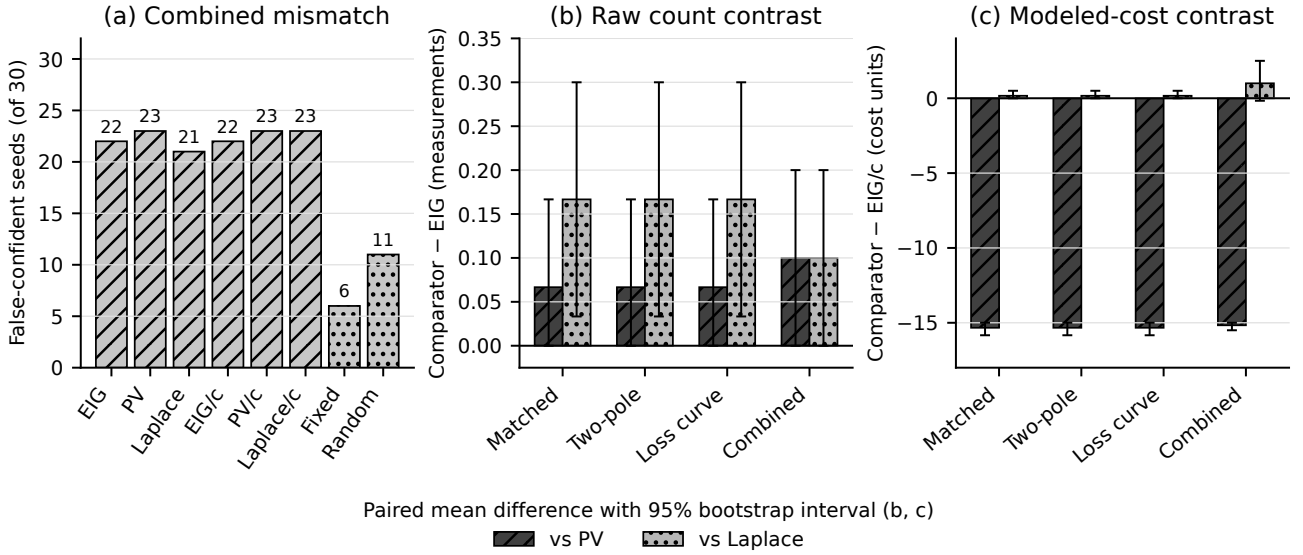


Fig. 4. Controlled-mismatch results. Bars and error bars show paired mean differences and descriptive 95% paired-bootstrap intervals without multiplicity adjustment; positive differences favor EIG. The /c policy suffix denotes prespecified modeled cost, not measured laboratory time. Combined-mismatch false-confidence counts are descriptive, not a causal comparison among policies.

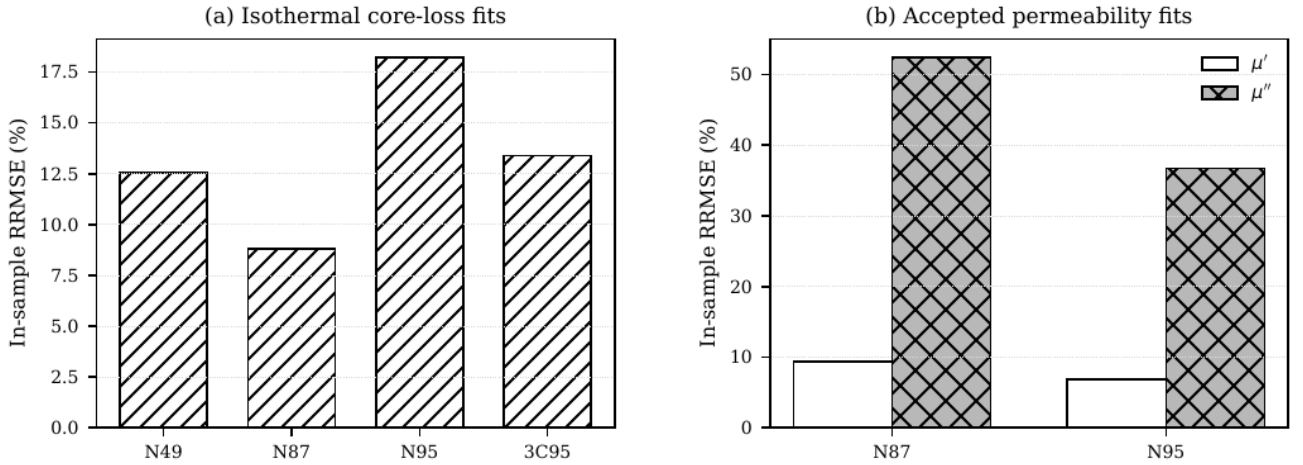


Fig. 5. In-sample measured-data adequacy, with storage and loss permeability reported separately.

paired seed under any of the three raw policies had passed both gates. Raw EIG still failed the L_m gate in 28 of 30 seeds; predictive variance still failed core loss in 16 and L_m in 14; Laplace D-optimality still failed core loss in 21 and L_m in 9. Each policy then completed the complementary response information at measurement five, so the integer count endpoint collapsed their different paths to the same value. Their ordered sequences confirm the distinction: EIG and predictive variance match in only 11 of 30 seeds, while EIG and Laplace D-optimality match in only 2 of 30. Trajectory source E5

The per-cost loss has a different mechanism. At the shared four-measurement state available in 22 paired seeds, the mean core-loss interval half-width is 26.03%, above its 8% gate, while the mean L_m half-width is 2.78%, below its 5% gate. EIG/cost nevertheless ranks an additional 10 kHz L_m measurement first in 21 of those states because its mean

joint information-per-cost utility is 0.05282, compared with 0.04390 for the high-leverage core-loss point. Predictive variance/cost ranks that core-loss point first in all 22 states and crosses both gates one measurement earlier. Trajectory source E5

Thus EIG/cost behaves consistently with its own objective, but that objective is not expected cost-to-gate. The experiment asks whether a global parameter- information utility is efficient for reaching two local predictive-width thresholds; it is not guaranteed to be. A gate-aligned utility must be preregistered and evaluated as a new experiment rather than substituted into this frozen result after observing the comparison.

9.3 Implications for converter modeling

For a converter workflow, the posterior is most useful as a conditional input to loss, inductance, and sensitivity calculations over the specified operating region. Posterior draws can be propagated through a component model instead of substituting one nominal coefficient set, allowing a designer to determine whether parameter uncertainty materially affects thermal or magnetic design margins. The Fisher spectrum can motivate testing whether a new channel or a wider, physically controlled excitation range would constrain a weak direction more effectively than densifying the original sweep [1], [28].

Measured residuals define a separate model-adequacy assessment. The accepted core-loss fits show the error remaining after estimating Steinmetz coefficients inside one temperature cohort. The much larger loss-component permeability residuals show that a single Cole–Cole pole cannot be treated as a broadband physical description for these records. A component calculation that is sensitive to μ'' should therefore retain an explicit discrepancy allowance or adopt a higher-order permeability law before relying on posterior parameter width. Conversely, a reasonable μ' residual does not validate μ'' , because the two components occupy different numerical scales and correspond to different design consequences.

The common evaluation protocol safeguards this interpretation by preventing combination of a favorable recovery run, a figure from another seed set, and a measured fit produced under different filters. Fixed model settings, data selections, and random seeds also make it possible to distinguish sampling variation from changes to the computational experiment.

9.4 Threats to validity

Internal validity is limited by five synthetic recovery seeds and 30 paired acquisition seeds. Pairing controls differences in candidate outcomes between policies. A prespecified convergence study compares candidate ranks and downstream endpoints against a higher-budget reference; finite nested Monte Carlo variance nevertheless remains. Synthetic recovery is matched-model, and the observed interval-inclusion count is too small to establish calibrated coverage.

The independent MM-3 seed set adds a controlled structural-mismatch test, but only for one two-pole permeability departure, one temperature/curvature loss departure, and their combination. Its 90% latent-holdout inclusion is a fixed-point summary, not simulation-based calibration. Production records retain sampler checkpoint histories but not full chains, so the stored convergence decision can be reconstructed while autocorrelation and acceptance cannot be independently recomputed without rerunning the locked source. Source E16

The policy comparison now includes deterministic and randomized balanced traversals, predictive variance, and Laplace D-optimality. It does not include a full nonlinear Fisher-greedy implementation, space-filling designs, multi-step look-ahead, or learned adaptive policies. More importantly, the direct results do not establish EIG dominance over the strong comparators. Observation noise and geome-

try are fixed, and estimator intervals are not total physical-uncertainty intervals.

Construct validity is limited by the low-order forward models and available data. The Steinmetz relation is temperature independent, and one Cole–Cole pole does not capture all ferrite relaxation, resonance, or loss processes [17]. Measured errors are in-sample. Public curves may also contain digitization, instrument, preprocessing, and source-specific errors not represented by the likelihood. Excluding convergence-invalid or boundary-active records prevents unsupported aggregate claims but restricts the population represented by the accepted subset.

External validity is confined to the specified channels, temperature cohorts, excitation ranges, materials, and cost model. The measured-data EIG ranking is a model-conditional acquisition suggestion, not a validated optimal laboratory plan, particularly given the observed μ'' discrepancy. Measurement-count reductions must not be interpreted as laboratory-time savings because setup, settling, calibration, rejection, and batch costs were not measured. The results do not replace held-out validation, controlled multi-lot characterization, switching-waveform experiments, or component qualification. Those extensions require new experimental evidence rather than broader interpretation of the present results.

The strong-comparator campaign already produced an important negative result: EIG did not beat either strong raw policy in the matched-model count comparison and lost the per-cost comparison with predictive variance. Publishing this outcome prevents the favorable fixed-traversal comparison from being generalized beyond its actual comparator. Sources E4 and E5

X. Conclusion

Bayesian calibration recovered the active Steinmetz and Cole–Cole parameters with per-parameter median absolute relative errors of 0.27%–5.85% across five matched-model synthetic data sets; the generating values lay within 28 of 30 model-conditional 90% credible intervals. Raw EIG reached the two-target precision gate in five measurements rather than nine for the fixed traversal, but it tied predictive variance and Laplace D-optimality in all 30 direct count comparisons. EIG/cost tied Laplace D-optimality and lost to predictive variance/cost by a mean 15.17 modeled-cost units. The accepted measured records produced substantially larger RRMSE for μ'' than for μ' , showing that posterior concentration cannot compensate for an inadequate permeability law. MM-3 reinforces that conclusion synthetically: combined-mismatch raw EIG reached the precision gate in every seed but was false-confident in 22 of 30, while its count advantage over either strong comparator was only 0.10 measurements. The local gate is therefore not evidence of truth accuracy under the locked structural departures. The defensible conclusion is therefore a reproducible model-conditional benchmark with informative positive and negative results, not proof of EIG superiority or laboratory efficiency. Evidence E2, E4, E7, and E16

XI. Exact Sequential Evaluation Logic

For each confirmatory seed, the synthetic truth is drawn once from the fixed prior. One outcome per exact candidate identity is then generated and stored. Each policy is evaluated by the following logic:

1. Start with the same declared P_v and μ' observations.
2. Fit the six-coordinate posterior and require all sampler gates.
3. If both target intervals satisfy the precision gate, retain count and modeled cost and stop successfully.
4. Otherwise score or traverse only unrevealed candidates. EIG uses $\hat{U}(d)$; predictive variance uses noise-normalized posterior response variance; Laplace D-optimality uses the log-determinant precision increment. Per-cost forms divide by $c(d)$. Fixed and randomized policies follow their declared channel-balanced traversals.
5. Break deterministic ties by exact design identity, reveal the stored outcome for the selected identity, append it, and return to step 2.
6. If the 25-measurement budget is exhausted, retain the trajectory as a failure rather than imputing a stopping count.

This logic separates three random objects: the prior-predictive truth, the candidate-indexed outcome table, and the policy's numerical scoring stream. Changing policy order changes neither the truth nor any candidate outcome. Stable design identities also make EIG scores invariant to permutation of the input library.

XII. Endpoint and Uncertainty Definitions

For a latent target $g(\theta)$, the reported precision statistic is

$$h_g = 100 \frac{q_{0.95}[g(\theta)] - q_{0.05}[g(\theta)]}{2|q_{0.50}[g(\theta)]|}.$$

The gate requires $h_{P_v} \leq 8$ and $h_{L_m} \leq 5$. The measurement-count endpoint is the number of revealed observations, including the two initial points, at the first satisfied gate. The cost endpoint is the sum of declared channel costs through that same step. Because the cost constants were not measured, the endpoint is dimensionless modeled burden even though the configuration names them as seconds for engineering convenience.

For paired seed s and a declared direct comparator, the endpoint difference is comparator minus EIG, so a positive value favors EIG. Resampling is performed over complete paired seeds, never over unpaired policy outcomes. Monte Carlo score intervals come from repeated estimates of the nested estimator; posterior intervals come from retained sampler draws; paired bootstrap intervals come from the 30 seed-level differences. These three intervals answer different questions and are not interchangeable.

XIII. Evidence and software availability

Numerical results are bound to validated scientific freeze 20260817T072230Z_401e3030fe13, manifest SHA-256 85448a2c3c9db2db051c94543d8a336e7157d55289f10c1792e9c57d433812f7. The freeze contains the 30 complete eight-policy trajectories, estimator states and scores, point-level 23-point holdouts, parameter-recovery records, and reconstructed endpoints. The sanitized v2 public audit bundle is available as a versioned release asset. The public bundle omits machine-specific operational metadata. Raw measured curves remain governed by the cited upstream sources and are not redistributed as if produced by this study. Release source E8

The separate MM-3 result is bound to aggregate SHA-256 03e8d81c48f3b5eb2c807a47b880972d3ea727b149788c2defc35f1ac1d222d. Its public audit asset contains all 120 sanitized task records and reconstructs the aggregate; it omits full walker-by-iteration chains. Model-mismatch source E16

The scientific job ledger accounts for every result artifact in the release, and Evidence Sources maps each quantitative result family to an exact pointer in the disclosure-safe projection.

References

- [1] K. Chaloner and I. Verdinelli, "Bayesian experimental design: A review," *Statistical Science*, vol. 10, no. 3, pp. 273–304, 1995, doi: 10.1214/ss/1177009939.
- [2] E. G. Ryan, C. C. Drovandi, J. M. McGree, and A. N. Pettitt, "A review of modern computational algorithms for bayesian optimal design," *International Statistical Review*, vol. 84, no. 1, pp. 128–154, 2016, doi: 10.1111/insr.12107.
- [3] C. P. Steinmetz, "On the law of hysteresis," *Transactions of the American Institute of Electrical Engineers*, vol. IX, no. 1, pp. 1–64, 1892, doi: 10.1109/T-AIEE.1892.5570437.
- [4] G. Bertotti, "General properties of power losses in soft ferromagnetic materials," *IEEE Transactions on Magnetics*, vol. 24, no. 1, pp. 621–630, 1988, doi: 10.1109/20.43994.
- [5] D. C. Jiles and D. L. Atherton, "Theory of ferromagnetic hysteresis," *Journal of Magnetism and Magnetic Materials*, vol. 61, no. 1–2, pp. 48–60, 1986, doi: 10.1016/0304-8853(86)90066-1.
- [6] J. Reinert, A. Brockmeyer, and R. W. A. D. Doncker, "Calculation of losses in ferro- and ferrimagnetic materials based on the modified steinmetz equation," *IEEE Transactions on Industry Applications*, vol. 37, no. 4, pp. 1055–1061, 2001, doi: 10.1109/28.936396.
- [7] J. Li, T. Abdallah, and C. R. Sullivan, "Improved calculation of core loss with nonsinusoidal waveforms," in *Conference Record of the 2001 IEEE Industry Applications Conference*, vol. 4, 2001, pp. 2203–2210, doi: 10.1109/IAS.2001.955931.
- [8] K. Venkatachalam, C. R. Sullivan, T. Abdallah, and H. Tacca, "Accurate prediction of ferrite core loss with nonsinusoidal waveforms using only steinmetz parameters," in *2002 IEEE Workshop on Computers in Power Electronics*, 2002, pp. 36–41, doi: 10.1109/CIPE.2002.1196712.
- [9] A. V. den Bossche, V. C. Valchev, and G. B. Georgiev, "Measurement and loss model of ferrites with non-sinusoidal waveforms," in *2004 IEEE 35th Annual Power Electronics Specialists Conference*, vol. 6, 2004, pp. 4814–4818. [Online]. Available: <https://hdl.handle.net/1854/LU-404490>
- [10] W. A. Roshen, "A practical, accurate and very general core loss model for nonsinusoidal waveforms," *IEEE Transactions on Power Electronics*, vol. 22, no. 1, pp. 30–40, 2007, doi: 10.1109/TPEL.2006.886608.

- [11] J. Muehlethaler, J. Biela, J. W. Kolar, and A. Ecklebe, "Improved core-loss calculation for magnetic components employed in power electronic systems," *IEEE Transactions on Power Electronics*, vol. 27, no. 2, pp. 964–973, 2012, doi: 10.1109/TPEL.2011.2162252.
- [12] S. Dobak, C. Beatrice, V. Tsakaloudi, and F. Fiorillo, "Magnetic losses in soft ferrites," *Magnetochemistry*, vol. 8, no. 6, p. 60, 2022, doi: 10.3390/magnetochemistry8060060.
- [13] E. C. Snelling, *Soft Ferrites: Properties and Applications*, 2nd ed. London, U.K.: Butterworths, 1988.
- [14] C. W. T. McLyman, *Transformer and Inductor Design Handbook*, 4th ed. CRC Press, 2011.
- [15] K. S. Cole and R. H. Cole, "Dispersion and absorption in dielectrics i. alternating current characteristics," *The Journal of Chemical Physics*, vol. 9, no. 4, pp. 341–351, 1941, doi: 10.1063/1.1750906.
- [16] J. L. Snoek, "Dispersion and absorption in magnetic ferrites at frequencies above one Mc/s," *Physica*, vol. 14, no. 4, pp. 207–217, 1948, doi: 10.1016/0031-8914(48)90038-X.
- [17] T. Tsutaoka, "Frequency dispersion of complex permeability in Mn–Zn and Ni–Zn spinel ferrites and their composite materials," *Journal of Applied Physics*, vol. 93, no. 5, pp. 2789–2796, 2003, doi: 10.1063/1.1542651.
- [18] H. Li, D. Serrano, T. Guilloid, E. Dogariu, A. Nadler, S. Wang, M. Luo, V. Bansal, Y. Chen, C. R. Sullivan, and M. Chen, "MagNet: An open-source database for data-driven magnetic core loss modeling," in *2022 IEEE Applied Power Electronics Conference and Exposition*, 2022, pp. 588–595, doi: 10.1109/APEC43599.2022.9773372.
- [19] Paderborn University, Power Electronics and Electrical Drives, "materialdatabase: Open material data for magnetic component design," Software and data repository, version 0.4.0, 2026, accessed 2026-08-05. [Online]. Available: <https://github.com/upb-lea/materialdatabase>
- [20] A. Gelman, J. B. Carlin, H. S. Stern, D. B. Dunson, A. Vehtari, and D. B. Rubin, *Bayesian Data Analysis*, 3rd ed. CRC Press, 2013.
- [21] J. Sacks, W. J. Welch, T. J. Mitchell, and H. P. Wynn, "Design and analysis of computer experiments," *Statistical Science*, vol. 4, no. 4, pp. 409–423, 1989, doi: 10.1214/ss/1177012413.
- [22] M. C. Kennedy and A. O'Hagan, "Bayesian calibration of computer models," *Journal of the Royal Statistical Society: Series B*, vol. 63, no. 3, pp. 425–464, 2001, doi: 10.1111/1467-9868.00294.
- [23] M. J. Bayarri, J. O. Berger, R. Paulo, J. Sacks, J. A. Cafeo, J. Cavendish, C.-H. Lin, and J. Tu, "A framework for validation of computer models," *Technometrics*, vol. 49, no. 2, pp. 138–154, 2007, doi: 10.1198/004017007000000092.
- [24] J. Brynjarsdottir and A. O'Hagan, "Learning about physical parameters: The importance of model discrepancy," *Inverse Problems*, vol. 30, no. 11, p. 114007, 2014, doi: 10.1088/0266-5611/30/11/114007.
- [25] R. Tuo and C. F. J. Wu, "Efficient calibration for imperfect computer models," *The Annals of Statistics*, vol. 43, no. 6, pp. 2331–2352, 2015, doi: 10.1214/15-AOS1314.
- [26] M. Plumlee, "Bayesian calibration of inexact computer models," *Journal of the American Statistical Association*, vol. 112, no. 519, pp. 1274–1285, 2017, doi: 10.1080/01621459.2016.1211016.
- [27] D. V. Lindley, "On a measure of the information provided by an experiment," *The Annals of Mathematical Statistics*, vol. 27, no. 4, pp. 986–1005, 1956, doi: 10.1214/aoms/1177728069.
- [28] A. C. Atkinson, A. N. Donev, and R. D. Tobias, *Optimum Experimental Designs, with SAS*. Oxford University Press, 2007, doi: 10.1093/oso/9780199296590.001.0001.
- [29] K. J. Ryan, "Estimating expected information gains for experimental designs with application to the random fatigue-limit model," *Journal of Computational and Graphical Statistics*, vol. 12, no. 3, pp. 585–603, 2003, doi: 10.1198/1061860032012.
- [30] T. Rainforth, R. Cornish, H. Yang, A. Warrington, and F. Wood, "On nesting monte carlo estimators," in *Proceedings of the 35th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 80, 2018, pp. 4267–4276. [Online]. Available: <https://proceedings.mlr.press/v80/rainforth18a.html>
- [31] T. Goda, T. Hironaka, and T. Iwamoto, "Multilevel monte carlo estimation of expected information gains," *Stochastic Analysis and Applications*, vol. 38, no. 4, pp. 581–600, 2020, doi: 10.1080/07362994.2019.1705168.
- [32] X. Huan and Y. M. Marzouk, "Simulation-based optimal bayesian experimental design for nonlinear systems," *Journal of Computational Physics*, vol. 232, no. 1, pp. 288–317, 2013, doi: 10.1016/j.jcp.2012.08.013.
- [33] A. Foster, M. Jankowiak, E. Bingham, P. Horsfall, Y. W. Teh, T. Rainforth, and N. Goodman, "Variational bayesian optimal experimental design," in *Advances in Neural Information Processing Systems*, vol. 32, 2019. [Online]. Available: <https://proceedings.neurips.cc/paper/2019/hash/d55cbf210f175f4a37916eafe6c04f0d-Abstract.html>
- [34] A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth, "Deep adaptive design: Amortizing sequential bayesian experimental design," in *Proceedings of the 38th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 139, 2021, pp. 3384–3395. [Online]. Available: <https://proceedings.mlr.press/v139/foster21a.html>
- [35] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948, doi: 10.1002/j.1538-7305.1948.tb01338.x.
- [36] S. Kullback and R. A. Leibler, "On information and sufficiency," *The Annals of Mathematical Statistics*, vol. 22, no. 1, pp. 79–86, 1951, doi: 10.1214/aoms/1177729694.
- [37] D. J. C. MacKay, "Information-based objective functions for active data selection," *Neural Computation*, vol. 4, no. 4, pp. 590–604, 1992, doi: 10.1162/neco.1992.4.4.590.
- [38] J. Kiefer and J. Wolfowitz, "The equivalence of two extremum problems," *Canadian Journal of Mathematics*, vol. 12, pp. 363–366, 1960, doi: 10.4153/CJM-1960-030-4.
- [39] J. Goodman and J. Weare, "Ensemble samplers with affine invariance," *Communications in Applied Mathematics and Computational Science*, vol. 5, no. 1, pp. 65–80, 2010, doi: 10.2140/camcos.2010.5.65.
- [40] D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman, "emcee: The MCMC hammer," *Publications of the Astronomical Society of the Pacific*, vol. 125, no. 925, pp. 306–312, 2013, doi: 10.1086/670067.
- [41] A. Gelman and D. B. Rubin, "Inference from iterative simulation using multiple sequences," *Statistical Science*, vol. 7, no. 4, pp. 457–472, 1992, doi: 10.1214/ss/1177011136.
- [42] A. Vehtari, A. Gelman, D. Simpson, B. Carpenter, and P.-C. Buerkner, "Rank-normalization, folding, and localization: An improved $\hat{R}$ for assessing convergence of MCMC (with discussion)," *Bayesian Analysis*, vol. 16, no. 2, pp. 667–718, 2021, doi: 10.1214/20-BA1221.
- [43] S. R. Cook, A. Gelman, and D. B. Rubin, "Validation of software for bayesian models using posterior quantiles," *Journal of Computational and Graphical Statistics*, vol. 15, no. 3, pp. 675–692, 2006, doi: 10.1198/106186006X136976.
- [44] S. Talts, M. Betancourt, D. Simpson, A. Vehtari, and A. Gelman, "Validating bayesian inference algorithms with simulation-based calibration," 2018, doi: 10.48550/arXiv.1804.06788.
- [45] A. Gelman, X.-L. Meng, and H. S. Stern, "Posterior predictive assessment of model fitness via realized discrepancies," *Statistica Sinica*, vol. 6, no. 4, pp. 733–807, 1996. [Online]. Available: <https://www3.stat.sinica.edu.tw/statistica/j6n4/j6n41/j6n41.htm>
- [46] J. Gabry, D. Simpson, A. Vehtari, M. Betancourt, and A. Gelman, "Visualization in bayesian workflow," *Journal of the Royal Statistical Society: Series A*, vol. 182, no. 2, pp. 389–402, 2019, doi: 10.1111/rssa.12378.
- [47] R. A. Fisher, "On the mathematical foundations of theoretical statistics," *Philosophical Transactions of the Royal Society A*, vol. 222, pp. 309–368, 1922, doi: 10.1098/rsta.1922.0009.
- [48] R. N. Gutenkunst, J. J. Waterfall, F. P. Casey, K. S. Brown, C. R. Myers, and J. P. Sethna, "Universally sloppy parameter sensitivities in systems biology models," *PLoS Computational Biology*, vol. 3, no. 10, p. e189, 2007, doi: 10.1371/journal.pcbi.0030189.
- [49] M. K. Transtrum, B. B. Machta, K. S. Brown, B. C. Daniels, C. R. Myers, and J. P. Sethna, "Perspective: Sloppiness and emergent theories in physics, biology, and beyond," *The Journal of Chemical Physics*, vol. 143, no. 1, p. 010901, 2015, doi: 10.1063/1.4923066.